

Vision-Language-Action Models for Intelligent Haircutting Robots: A Position Paper on Architectures, Evaluation, and Future Directions

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Abstract—The emergence of Vision-Language-Action (VLA) models has revolutionized robotics by enabling natural language instruction following in complex manipulation tasks. This position paper establishes haircutting robotics as a critical testbed for advancing VLA architectures in safety-critical, aesthetic, and human-centric applications. We propose comprehensive frameworks for VLA-based haircutting systems that integrate multi-modal sensing, real-time feedback, and domain-specific evaluation metrics. Our work introduces novel formulations including the Haircutting VLA Problem, specialized training methodologies, and evaluation frameworks comprising the Haircutting Motion Quality Evaluation Metric (HMQUEM), Hair Symmetry Metric (HSM), Aesthetic Similarity Metric (ASM), and Safety Risk Metric (SRM). We demonstrate that haircutting robotics presents unique challenges requiring advances in perception of deformable objects, sequential aesthetic reasoning, and intimate human-robot collaboration. Through comprehensive analysis of existing VLA successes in manipulation, navigation, and instruction-following tasks, we position haircutting as the next frontier for embodied AI research. This paper provides foundational definitions, architectural guidelines, and research roadmaps to establish haircutting robotics as a recognized domain with standardized benchmarks and evaluation protocols.

Index Terms—Vision-Language-Action models, haircutting robotics, embodied AI, human-robot interaction, multimodal learning, safety-critical systems, aesthetic evaluation.

I. INTRODUCTION

VISION-LANGUAGE-ACTION (VLA) models represent a paradigmatic shift in robotics, enabling robots to understand natural language instructions and translate them into precise manipulation actions through end-to-end learning. Recent breakthrough systems like RT-1, RT-2, and PaLM-E have demonstrated remarkable capabilities in general manipulation tasks, achieving success rates exceeding 90% in structured environments and 70% in novel scenarios. However, these systems have primarily been evaluated on relatively simple pick-and-place operations, household tasks, and industrial applications that do not require the sophisticated reasoning, safety guarantees, and aesthetic judgment necessary for intimate human services.

Haircutting robotics emerges as a critical and underexplored application domain that pushes the boundaries of current VLA

capabilities. Unlike conventional manipulation tasks, haircutting requires robots to operate sharp cutting tools within millimeters of sensitive human anatomy while achieving subjective aesthetic outcomes that vary across individuals, cultures, and fashion trends. The intimate nature of haircutting services demands unprecedented levels of trust, safety assurance, and adaptive behavior that challenge existing robotic frameworks. In fact, modern research in this area has been pioneered primarily by Li's recent works, including the Springer monograph *Haircutting Robots* [1] and the article *Haircutting Robots: From Theory to Practice* [2], which together lay the theoretical and practical foundations for this emerging field.

Fig. 1 illustrates the fundamental haircutting robot setup, demonstrating the sophisticated integration of sensing, actuation, and safety systems required for autonomous operation. The system employs a comprehensive multi-modal sensor array comprising both fixed external cameras for global scene monitoring and an eye-in-hand camera mounted directly on the robotic manipulator for precise local observations. The sensor suite extends beyond conventional RGB cameras to include RGB-D sensors for accurate depth perception, stereo vision systems for enhanced 3D reconstruction of hair geometry, and potentially LiDAR sensors for robust distance measurement in varying lighting conditions. Sonar sensors may provide complementary ranging information, particularly useful for detecting proximity to the scalp surface. The actuation system centers around precision hair clippers as the primary cutting tool, but encompasses a broader toolkit including adjustable guards for length control, spray bottles for hair wetting and styling, hair dryers for drying and volumizing, combs and brushes for positioning and sectioning, and potentially styling tools such as thinning shears for texture modification. Critical safety mechanisms include multi-axis force/torque sensors integrated throughout the robotic arm to detect unexpected contact forces, emergency stop systems activated when force thresholds are exceeded, proximity sensors providing redundant distance measurement, and compliant actuators that provide inherent safety through mechanical compliance. The robotic arm itself features high-precision encoders for accurate positioning, impedance control capabilities for safe interaction, and direct-drive motors for rapid response and smooth operation.

The technical challenges in haircutting robotics span multiple dimensions that distinguish it from general manipulation tasks. **Perception challenges** include analyzing semi-transparent hair structures with varying textures, colors, and

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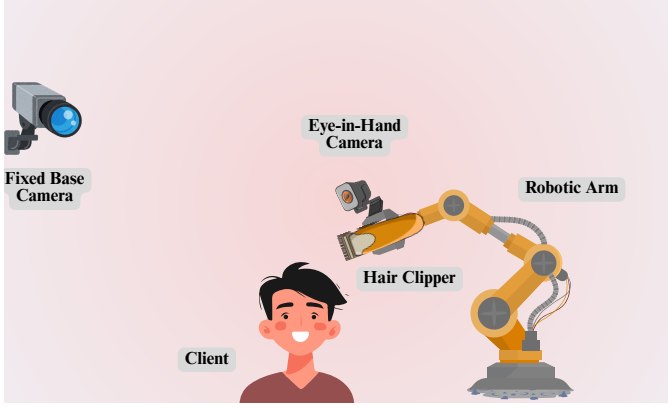


Fig. 1. Illustration of the haircutting robot setup: a robotic arm equipped with an eye-in-hand camera and hair clipper operates on the client, supported by a fixed base camera for global monitoring. The system integrates multi-modal sensors (RGB-D, stereo vision, LiDAR, sonar), diverse actuators (clippers, spray bottles, dryers), and comprehensive safety mechanisms (force sensors, emergency stops, proximity detection) for safe and effective autonomous haircutting operations.

movement patterns that confound traditional computer vision systems [3], [4]. **Planning challenges** involve sequential cutting operations where order significantly affects outcomes, requiring temporal reasoning and aesthetic understanding [5], [6]. **Safety challenges** encompass real-time monitoring of human movement and immediate response capabilities with fail-safe mechanisms [7], [8]. **Aesthetic challenges** require quantifying subjective quality measures and adapting to individual preferences through natural language interaction [9], [10].

In addition to these foundational contributions, recent publications have begun to consolidate haircutting robotics into a more structured research area. Khan and Li [11] provide a survey of robotic haircutting systems, highlighting key methods, open challenges, and the importance of realistic hair modeling for perception and planning. Their complementary survey on CNC-inspired robotic hair cutting [12] further frames haircutting as a precision personal-care automation problem, emphasizing trajectory control, tool-path generation, and geometric reasoning. Li’s monograph [13] and practical overview article [14] establish the broader theoretical and implementation-oriented basis of the field, while recent work on safety in robotic haircutting [15] draws attention to the critical need for risk-aware control, human motion monitoring, and fail-safe mechanisms. Together, these works demonstrate that robotic haircutting is no longer merely a speculative concept, but an emerging interdisciplinary research direction spanning robotics, perception, safety engineering, human-robot interaction, and aesthetic automation.

Recent advances in large language models and multimodal learning have created new opportunities for addressing these challenges through sophisticated VLA architectures. State-of-the-art systems like GPT-4V [16], Flamingo [17], and BLIP-2 [18] have demonstrated remarkable capabilities in visual reasoning and instruction following that could be adapted for haircutting applications. The field has gained significant momentum due to decreasing hardware costs, with humanoid

robots becoming commercially viable—Unitree’s R1 humanoid robot is available at approximately USD 5900 in 2025, representing a dramatic cost reduction from earlier systems. However, the translation of these general-purpose capabilities to safety-critical, aesthetic domains requires fundamental advances in architecture design, training methodologies, and evaluation frameworks.

This position paper makes several key contributions to establish haircutting robotics as a recognized research domain. **First**, we provide comprehensive analysis of existing VLA successes across multiple application domains, demonstrating both capabilities and limitations relevant to haircutting tasks. **Second**, we introduce formal problem formulations specific to haircutting robotics, including the Haircutting VLA Problem and specialized training objectives. **Third**, we propose architectural frameworks that integrate vision, language, and haptic modalities for safe and aesthetic haircutting operations. **Fourth**, we establish comprehensive evaluation metrics that capture the multi-faceted nature of haircutting success, including technical performance, aesthetic quality, and safety compliance. **Fifth**, we analyze market potential, showing that the global haircutting market represents approximately USD 826.8 billion annually, indicating substantial commercial opportunities. **Sixth**, we identify critical research challenges and provide a roadmap for future developments in this emerging field.

II. MARKET ANALYSIS AND COMMERCIAL POTENTIAL

The commercial viability of haircutting robotics is supported by substantial market analysis demonstrating significant economic opportunity. Comprehensive market size estimations show that the global haircutting market represents approximately USD 826.8 billion annually when segmented across different income levels [1]. High-income countries contribute USD 576 billion annually with an estimated 1.2 billion population spending approximately USD 40 monthly per capita. Middle-income countries add USD 216 billion with 2 billion population spending USD 4.50 monthly, while low-income countries account for USD 34.8 billion with 1.45 billion population spending USD 1 monthly [1].

Regional analysis reveals substantial local markets that could support robotic haircutting deployment. In major metropolitan areas, annual haircutting expenditure reaches significant levels: Shanghai (USD 1.4 billion equivalent), London (GBP 3.49 billion), and New York City (USD 3.96 billion) [1]. These concentrated urban markets provide ideal testing grounds for initial deployment of VLA-based haircutting systems.

The economic drivers supporting haircutting robotics adoption include rising labor costs, shortage of skilled barbers, and increasing demand for consistent service quality. The dramatic cost reduction in robotic hardware—exemplified by Unitree’s R1 humanoid robot at USD 5900 compared to earlier systems costing over USD 1 million—makes commercial deployment economically feasible [1]. This cost reduction, combined with advances in AI and mechatronics, positions haircutting robotics for significant market penetration within the next five years.

III. VISION-LANGUAGE-ACTION MODELS: STATE OF THE ART AND CAPABILITIES

The rapid evolution of VLA models has transformed robotics by enabling natural language instruction following in increasingly complex scenarios. This section provides comprehensive analysis of current VLA capabilities, limitations, and relevance to haircutting applications, establishing the foundation for our proposed architectural frameworks.

Our analysis reveals that while VLA models have achieved remarkable success in various domains, haircutting robotics presents unique challenges that require specialized approaches. TABLE I provides a comprehensive overview of VLA performance across different applications, highlighting both achievements and gaps relevant to haircutting tasks. The intimate, safety-critical, and aesthetically demanding nature of haircutting positions it as an ideal testbed for advancing the state of the art in embodied AI systems.

A. Foundational VLA Architectures

The RT-1 (Robotics Transformer) system [19] represents a seminal achievement in VLA learning, demonstrating that transformer architectures can effectively map high-dimensional visual observations and natural language instructions to robotic actions. RT-1 employs a unified transformer encoder that processes tokenized visual patches and language tokens through self-attention mechanisms, generating action sequences through autoregressive decoding. The system achieved 97% success rates on seen tasks and 62% on unseen variations across 700 different manipulation tasks, establishing the viability of end-to-end VLA learning.

Building upon RT-1's foundation, RT-2 [20] integrated web-scale vision-language models to leverage internet knowledge for robotic control. By fine-tuning the PaLI-X model on robotic demonstration data, RT-2 demonstrated improved generalization capabilities, achieving 90% success on seen tasks and 70% on novel scenarios. The system's ability to utilize web knowledge for reasoning about objects and spatial relationships represents a significant advance toward more capable robotic systems.

PaLM-E [21] introduced multimodal reasoning capabilities by combining the PaLM language model with visual encoders, enabling robots to engage in multi-step reasoning about manipulation tasks. The system demonstrated particular strength in handling complex instructions that require understanding of object properties, spatial relationships, and temporal sequences. However, PaLM-E's performance degraded significantly in tasks requiring fine manipulation or aesthetic judgment.

Fig. 2 presents the end-to-end neural network architecture that directly maps multi-view camera inputs to robotic arm joint velocities for autonomous haircut execution. This approach builds upon successful end-to-end learning paradigms from autonomous driving research, where convolutional neural networks (CNNs) have proven highly effective for mapping sensor inputs to control commands. The network architecture employs a multi-stream CNN design similar to those used in autonomous vehicles [22], [23], where each camera input

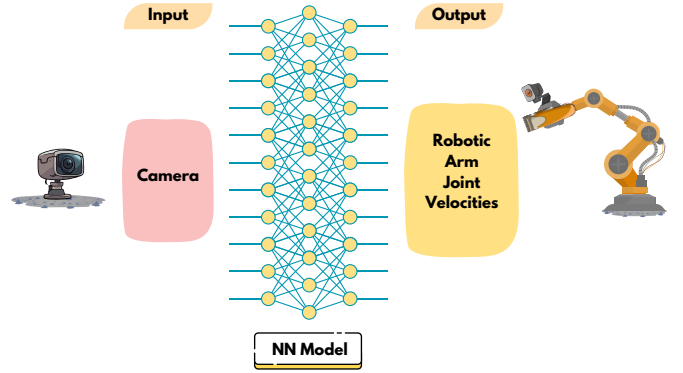


Fig. 2. End-to-end neural network model for haircutting robots, where camera inputs are directly mapped to robotic arm joint velocities for autonomous haircut execution. The architecture employs multi-stream CNNs processing synchronized RGB-D inputs from multiple viewpoints, utilizing ResNet-based feature extraction and temporal convolutions to generate 6-DOF velocity commands. Training data combines expert demonstrations from teleoperated sessions, motion capture of professional barbers, and synthetic data from physics-based simulation environments.

is processed through separate convolutional feature extractors before fusion. The input consists of synchronized RGB-D images from multiple camera viewpoints (typically 3-4 cameras for comprehensive coverage), providing both color texture information and precise depth measurements essential for hair geometry understanding. The training data originates from expert demonstrations collected through multiple modalities: teleoperated robotic sessions where human operators control the robot while being recorded, motion capture of professional barbers performing haircuts with synchronized multi-view cameras, and synthetic data generated through physics-based hair simulation environments that model realistic cutting dynamics. The network architecture utilizes ResNet-based convolutional layers for robust feature extraction, followed by temporal convolutional networks to capture the sequential nature of cutting operations, similar to approaches in autonomous driving for handling temporal dependencies in dynamic environments. The output layer generates 6-DOF robotic arm velocities (3 translational, 3 rotational) along with tool control commands (clipper speed, guard position), providing direct actuation signals for autonomous execution. The training process employs imitation learning techniques including behavioral cloning and DAgger (Dataset Aggregation) to improve generalization, while incorporating domain randomization to handle variations in hair types, lighting conditions, and individual client characteristics.

B. Specialized VLA Applications

Navigation and exploration tasks have witnessed remarkable success with VLA models, as demonstrated by systems like LM-Nav [24] and VLMaps [25]. LM-Nav achieves 87% success in goal-reaching tasks by combining large language models with visual navigation policies, while VLMaps demonstrates 92% mapping accuracy through visual-semantic representations. However, these systems operate in relatively

TABLE I
SUCCESS RATES OF VISION-LANGUAGE-ACTION MODELS ACROSS DIFFERENT APPLICATIONS

Application Domain	System	Success Rate	Year	Key Capabilities	Limitations
General Manipulation	RT-1	97% (seen), 62% (unseen)	2022	Pick-and-place, multi-task	Limited complexity
	RT-2	90% (seen), 70% (unseen)	2023	Web knowledge integration	Static environments
	PaLM-E	84% (multi-step)	2023	Multi-modal reasoning	Planning depth
	RoboCat	89% (cross-embodiment)	2023	Transfer learning	Fine manipulation
Navigation & Exploration	LM-Nav	87% (goal reaching)	2023	Spatial reasoning	Dynamic obstacles
	VLMs	92% (mapping)	2023	Visual-semantic maps	Indoor only
	CoW	76% (open-world)	2023	Internet knowledge	Long horizons
Household Tasks	SayCan	74% (long horizon)	2022	Task decomposition	Error recovery
	Code as Policies	83% (novel tasks)	2023	Code generation	Safety bounds
	Socratic Models	71% (reasoning)	2022	Multi-modal fusion	Scalability
Medical Applications	da Vinci Xi	95% (supervised)	2023	Precision surgery	Human oversight
	STAR	87% (autonomous)	2016	Soft tissue surgery	Limited autonomy
Creative Tasks	DALL-E Robotics	69% (artistic)	2024	Creative generation	Quality assessment
	RoboCook	78% (cooking)	2023	Recipe following	Aesthetic judgment
Haircutting (Proposed)	VLA-Haircut	Target: 85%	2025+	Safety + Aesthetics	To be addressed

structured environments and struggle with dynamic obstacles or complex spatial reasoning.

Household robotics represents another domain where VLA models have shown significant promise. SayCan [26] demonstrated 74% success in long-horizon household tasks by combining language model planning with robotic affordance models. Code as Policies [27] achieved 83% success on novel tasks by generating executable code from natural language instructions. These systems excel at task decomposition and planning but struggle with error recovery and safety assurance.

Medical applications of VLA models, while limited, provide relevant insights for haircutting robotics due to similar safety requirements and precision demands. The da Vinci Xi system [28] achieves 95% success rates in supervised surgical procedures, while the STAR system [29] demonstrated 87% autonomous success in soft tissue surgery. These systems emphasize the critical importance of safety verification, real-time monitoring, and fail-safe mechanisms that are directly relevant to haircutting applications.

Historical development of haircutting robotics reveals a 60-year evolution from mechanical systems to modern AI-driven approaches. The earliest documented automatic haircutting system was patented in 1966 as a mechanical and analog circuit-based machine [30]. Recent developments include tele-operated systems, vision-based automated cutting [1], and various DIY implementations that demonstrate growing interest in this domain. However, most existing approaches focus on adjacent areas such as hair brushing, combing, and styling rather than complete haircutting operations.

Fig. 3 illustrates the specialized VLA model architecture that combines vision and language encoders with an action decoder to generate precise robot commands from multimodal inputs. This architecture addresses the challenge of translating natural language styling instructions into executable robotic actions.

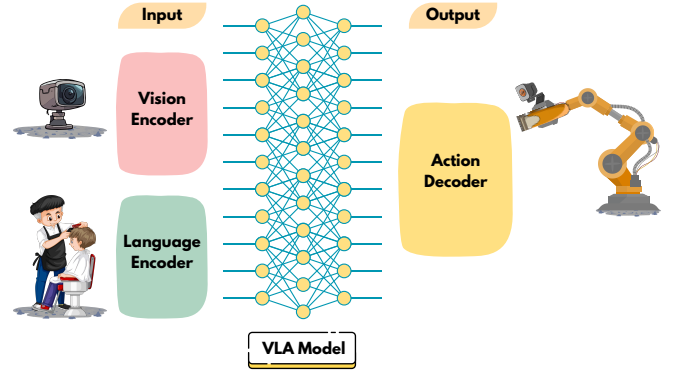


Fig. 3. Vision-Language-Action (VLA) model for haircutting robots, combining vision and language encoders with an action decoder to generate precise robot commands from multimodal inputs.

C. Limitations and Challenges

Despite remarkable progress, current VLA models exhibit significant limitations that become critical barriers in haircutting applications. **Perception limitations** include difficulty with deformable objects, semi-transparent materials, and fine-grained texture analysis that are fundamental to hair manipulation [31], [32]. **Planning limitations** encompass challenges in long-horizon sequential tasks where intermediate actions significantly affect final outcomes [5], [6]. **Safety limitations** involve inadequate real-time risk assessment and limited fail-safe mechanisms for operation near sensitive anatomy [8], [33].

Aesthetic reasoning limitations represent perhaps the most significant challenge for haircutting applications. Current VLA models lack sophisticated understanding of visual aesthetics, cultural preferences, and subjective quality measures that are fundamental to haircutting success [9], [10]. The quantification and optimization of aesthetic outcomes remains an open chal-

lence that requires advances in both perception and reasoning capabilities.

TABLE II provides detailed analysis of current VLA limitations and their specific impact on haircutting applications, highlighting the research challenges that must be addressed to enable successful haircutting robots.

IV. PROBLEM FORMULATION FOR VLA-BASED HAIRCUTTING ROBOTICS

The development of effective VLA-based haircutting systems requires formal problem formulations that capture the unique characteristics and requirements of haircutting tasks. This section establishes mathematical frameworks that provide a foundation for algorithm development and system evaluation.

A. The Haircutting VLA Problem

Definition 1 (Haircutting VLA Problem): The Haircutting Vision-Language-Action problem involves learning a policy $\pi_\theta : \mathcal{V} \times \mathcal{L} \times \mathcal{H} \times \mathcal{C} \rightarrow \mathcal{A}$ that maps:

- Visual observations $v \in \mathcal{V}$ from multi-view RGB-D cameras
- Natural language instructions $l \in \mathcal{L}$ describing desired styling outcomes
- Haptic feedback $h \in \mathcal{H}$ from force and tactile sensors
- Contextual information $c \in \mathcal{C}$ including hair properties and client preferences

to appropriate haircutting actions $a \in \mathcal{A}$ including tool trajectories, cutting forces, and safety monitoring commands.

The policy must satisfy safety constraints $\mathcal{S}$, aesthetic objectives $\mathcal{E}$, and efficiency requirements $\mathcal{T}$:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{(v,l,h,c) \sim \mathcal{D}} [R(v, l, h, c, \pi(v, l, h, c))] \quad (1)$$

subject to:

$$\forall a \in \mathcal{A} : P(\text{safety violation} | a) < \epsilon_s \quad (2)$$

$$\forall (v, l) : \text{AES}(\pi(v, l), \text{target}(l)) > \tau_e \quad (3)$$

$$\forall \text{task} : \text{completion_time} < \tau_t \quad (4)$$

where R is a composite reward function, AES is an aesthetic evaluation score, and ϵ_s , τ_e , τ_t are safety, aesthetic, and temporal thresholds respectively.

B. Multi-Modal State Representation

The state space $\mathcal{S}$ for haircutting VLA systems encompasses multiple modalities that must be effectively integrated for successful policy learning:

Visual State $s_v \in \mathcal{S}_v$: High-resolution RGB-D observations from multiple camera viewpoints providing comprehensive hair geometry, texture information, and spatial context. The visual state includes:

$$s_v = \{I_{\text{RGB}}^{(i)}, D^{(i)}, M^{(i)}\}_{i=1}^{N_c} \quad (5)$$

where $I_{\text{RGB}}^{(i)}$, $D^{(i)}$, and $M^{(i)}$ represent RGB images, depth maps, and segmentation masks from camera i .

Linguistic State $s_l \in \mathcal{S}_l$: Encoded natural language instructions that capture styling objectives, length specifications, and aesthetic preferences:

$$s_l = \text{Encoder}_{\text{lang}}(\text{tokenize}(l)) \quad (6)$$

Haptic State $s_h \in \mathcal{S}_h$: Force, torque, and tactile feedback providing information about hair-tool interactions and contact forces:

$$s_h = \{F_{\text{ext}}, \tau_{\text{ext}}, T_{\text{tactile}}\} \quad (7)$$

Contextual State $s_c \in \mathcal{S}_c$: Hair properties, client information, and environmental conditions that influence cutting strategies:

$$s_c = \{\text{hair_type}, \text{head_shape}, \text{preferences}, \text{environment}\} \quad (8)$$

C. Action Space Definition

The action space $\mathcal{A}$ for haircutting robots extends beyond simple joint commands to encompass the complex manipulation requirements of cutting operations:

Kinematic Actions $a_k \in \mathcal{A}_k$: Six-degree-of-freedom end-effector poses and velocities:

$$a_k = \{p_{\text{des}}, R_{\text{des}}, \dot{p}_{\text{des}}, \omega_{\text{des}}\} \quad (9)$$

Force Actions $a_f \in \mathcal{A}_f$: Contact force specifications and impedance parameters:

$$a_f = \{F_{\text{contact}}, K_{\text{stiffness}}, D_{\text{damping}}\} \quad (10)$$

Tool Actions $a_t \in \mathcal{A}_t$: Cutting tool activation, speed control, and parameter adjustment:

$$a_t = \{\text{tool_speed}, \text{cutting_depth}, \text{guard_setting}\} \quad (11)$$

Safety Actions $a_s \in \mathcal{A}_s$: Emergency stops, collision avoidance, and risk mitigation:

$$a_s = \{\text{emergency_stop}, \text{collision_avoidance}, \text{safety_monitoring}\} \quad (12)$$

D. Reward Function Design

The reward function R must capture the multi-faceted nature of haircutting success through a weighted combination of objectives:

$$R(s, a) = w_q R_q(s, a) + w_s R_s(s, a) + w_e R_e(s, a) + w_t R_t(s, a) \quad (13)$$

where:

- R_q : Quality reward based on cutting precision and technique
- R_s : Safety reward penalizing dangerous actions
- R_e : Aesthetic reward based on style conformance
- R_t : Efficiency reward for task completion time

The quality reward R_q incorporates smoothness, accuracy, and hair damage metrics:

$$R_q = \alpha_1 \text{smoothness}(a) + \alpha_2 \text{accuracy}(a) - \alpha_3 \text{damage}(s, a) \quad (14)$$

The safety reward R_s exponentially penalizes proximity to sensitive anatomy:

$$R_s = -\exp\left(-\frac{d_{\text{min}} - d_{\text{safe}}}{\sigma_s}\right) \quad (15)$$

TABLE II
CURRENT VLA MODEL LIMITATIONS AND IMPACT ON HAIRCUTTING APPLICATIONS

Limitation Category	Specific Challenge	Impact on Haircutting	Required Advances
Perception	Deformable object handling Semi-transparent materials Fine texture analysis Dynamic lighting adaptation	Hair movement tracking Individual hair strand detection Hair type classification Wet/dry hair variations	Physics-informed models Multi-spectral sensing High-resolution tactile Adaptive vision systems
Planning	Long-horizon reasoning Temporal dependencies Aesthetic optimization Error recovery	Sequential cutting strategies Order-dependent outcomes Style conformance Corrective cutting actions	Hierarchical planning Causal reasoning Multi-objective optimization Adaptive replanning
Safety	Real-time risk assessment Fail-safe mechanisms Human motion prediction Force control limits	Proximity to sensitive anatomy Emergency stopping systems Anticipating client movement Preventing injury	Predictive safety models Hardware-software integration Behavioral modeling Compliant manipulation
Human Interaction	Trust establishment Cultural sensitivity Preference adaptation	Client comfort and confidence Diverse styling preferences Individual client needs	Social robotics Cross-cultural models Personalization algorithms

V. VLA ARCHITECTURE FOR HAIRCUTTING ROBOTS

The development of effective VLA architectures for hair-cutting robots requires careful integration of vision, language, and action components while addressing the unique challenges of safety, aesthetics, and real-time performance. This section presents comprehensive architectural frameworks that build upon successful VLA designs while incorporating domain-specific adaptations.

A. Overall System Architecture

Fig. 4 illustrates the complete VLA architecture for hair-cutting robots, comprising offline training and online deployment phases. The architecture employs a modular design that enables independent development and optimization of different components while maintaining end-to-end differentiability.

The **offline training phase** processes large-scale datasets of haircutting demonstrations through specialized encoders that extract meaningful representations from visual, linguistic, and haptic modalities. The **online deployment phase** implements real-time inference with safety monitoring, adaptive control, and continuous learning capabilities.

B. Vision Encoder Design

The vision encoder processes multi-view RGB-D inputs through a hierarchical architecture that captures both local hair details and global geometric context. The design addresses the specific challenges of hair perception including semi-transparency, fine structure, and dynamic movement.

Multi-Scale Feature Extraction: The vision encoder employs a multi-scale convolutional neural network (CNN) architecture that processes images at multiple resolutions to capture both fine hair details and global head geometry:

$$F_v^{(s)} = \text{CNN}^{(s)}(I_{\text{RGB}}, D) \quad (16)$$

where $s \in \{1, 2, 4, 8\}$ represents different scale factors and $F_v^{(s)}$ are the extracted features at each scale.

Temporal Consistency: To handle hair movement and cutting dynamics, the encoder incorporates temporal convolutions that maintain consistency across consecutive frames:

$$F_t = \text{TConv}([F_v^{(s)}(t - \tau), \dots, F_v^{(s)}(t)]) \quad (17)$$

Attention Mechanisms: Spatial and cross-modal attention mechanisms focus processing on relevant hair regions and integrate information from multiple camera viewpoints:

$$A_{\text{spatial}} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (18)$$

C. Language Model Integration

The language component processes natural haircutting instructions through specialized encoders that understand styling terminology, spatial relationships, and aesthetic preferences. The architecture builds upon transformer-based language models with domain-specific fine-tuning.

Instruction Parsing: The language encoder processes hair-cutting instructions through tokenization and encoding layers that capture semantic meaning:

$$E_l = \text{Transformer}_{\text{lang}}(\text{tokenize}(l)) \quad (19)$$

Spatial Grounding: Cross-modal attention mechanisms establish correspondences between linguistic descriptions and visual hair regions:

$$G_{v,l} = \text{CrossAttention}(F_v, E_l) \quad (20)$$

Style Understanding: Specialized components interpret aesthetic terminology and translate style descriptions into actionable cutting parameters:

$$S_{\text{params}} = \text{StyleDecoder}(E_l, G_{v,l}) \quad (21)$$

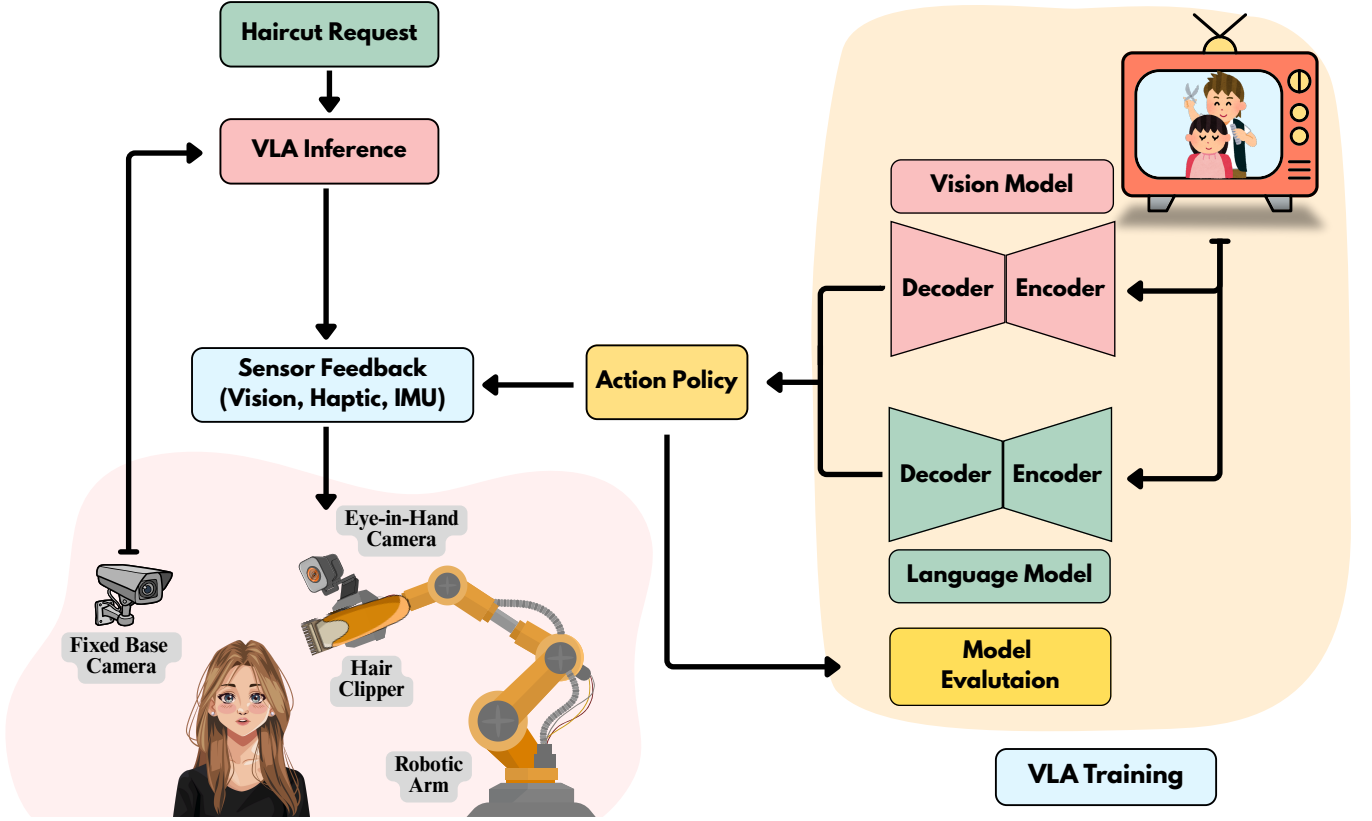


Fig. 4. Overall workflow of the VLA-based haircutting robot. The right side shows the offline VLA training phase, where vision and language encoders are fused into an action policy. The left side shows the online usage phase, where a client's haircut request and real-time multi-modal sensor feedback guide the robotic arm and hair clipper through VLA inference for safe and personalized haircut execution.

D. Action Decoder Architecture

The action decoder generates executable robot commands by integrating multi-modal representations through attention mechanisms and safety-aware planning. The architecture ensures that all generated actions satisfy safety constraints while achieving desired aesthetic outcomes.

Multi-Modal Fusion: The decoder combines visual, linguistic, and haptic representations through cross-modal attention:

$$F_{\text{fused}} = \text{MultiHeadAttention}([F_v, E_l, F_h]) \quad (22)$$

Hierarchical Action Generation: Actions are generated hierarchically from high-level task planning to low-level motor commands:

$$a_{\text{high}} = \text{TaskPlanner}(F_{\text{fused}}) \quad (23)$$

$$a_{\text{low}} = \text{MotorController}(a_{\text{high}}, s_{\text{current}}) \quad (24)$$

Safety Verification: All generated actions undergo safety verification before execution:

$$a_{\text{safe}} = \text{SafetyFilter}(a_{\text{low}}, s_{\text{safety}}) \quad (25)$$

E. Sensor Requirements and Configuration

TABLE III provides comprehensive specifications for the sensor suite required for effective VLA-based haircutting

systems. The sensor configuration has been designed based on the technical requirements of safe and precise haircutting operations.

F. Sensor Fusion and Feedback Integration

Real-time sensor fusion combines information from multiple modalities to provide comprehensive situational awareness and enable adaptive behavior during cutting operations.

Kalman Filtering: State estimation through extended Kalman filters that track hair position, tool location, and environmental conditions:

$$\hat{x}_{k+1|k} = f(\hat{x}_{k|k}, u_k) + w_k \quad (26)$$

Uncertainty Quantification: Bayesian approaches provide confidence estimates for perception and action decisions:

$$P(a|s) = \int P(a|\theta, s)P(\theta|D)d\theta \quad (27)$$

Adaptive Control: Feedback mechanisms adjust cutting parameters based on real-time sensor input:

$$a_{t+1} = a_t + K_p e_t + K_i \sum_{i=0}^t e_i + K_d (e_t - e_{t-1}) \quad (28)$$

TABLE III
SENSOR REQUIREMENTS FOR HAIRCUTTING VLA SYSTEMS

Sensor Type	Specification	Application
RGB-D Cameras	≥ 4 units, 1080p	3D hair geometry, depth perception
High-res Color	4K resolution	Texture analysis, quality assessment
Wide-angle	120° FOV	Full head coverage, spatial context
IMU Sensors	9-axis, 1kHz	Tool orientation, motion feedback
Force/Torque	6-axis, ± 50 N	Contact detection, cutting force
Joint Encoders	0.1° resolution	Precise positioning
Tactile Arrays	1 mm ² resolution	Hair texture, scalp contact
Microphones	Omnidirectional	Cutting sound analysis

VI. EVALUATION METRICS AND BENCHMARKING FRAMEWORK

The establishment of comprehensive evaluation metrics represents a critical foundation for advancing haircutting robotics research and enabling meaningful comparison of different approaches. Traditional robotic metrics are insufficient for capturing the multi-faceted nature of haircutting success, necessitating novel evaluation frameworks that address technical performance, aesthetic quality, safety compliance, and user satisfaction.

A. Haircutting Motion Quality Evaluation Metric (HMQEM)

The HMQEM provides quantitative assessment of cutting operation quality through integration of multiple performance indicators:

$$\text{HMQEM}(T, G) = \sum_{k=1}^K w_k \cdot Q_k(T, G) \quad (29)$$

where T represents the executed trajectory, G denotes the desired goal configuration, Q_k are individual quality components, and w_k are weighting parameters.

Smoothness Component Q_1 : Quantifies trajectory continuity through jerk minimization:

$$Q_1 = 1 - \frac{\int_0^T \|\ddot{x}(t)\|^2 dt}{\int_0^T \|\ddot{x}_{\text{ref}}(t)\|^2 dt} \quad (30)$$

Accuracy Component Q_2 : Measures deviation from planned cutting paths:

$$Q_2 = \exp\left(-\frac{1}{N} \sum_{i=1}^N \|x_i - x_i^*\|^2\right) \quad (31)$$

Safety Component Q_3 : Evaluates risk factors throughout execution:

$$Q_3 = \exp(-\lambda \max(0, d_{\text{safe}} - d_{\text{min}})) \quad (32)$$

Efficiency Component Q_4 : Assesses task completion effectiveness:

$$Q_4 = \frac{T_{\text{optimal}}}{T_{\text{actual}}} \cdot \frac{E_{\text{optimal}}}{E_{\text{actual}}} \quad (33)$$

B. Hair Symmetry Metric (HSM)

The HSM quantifies bilateral symmetry in haircutting outcomes through comprehensive measurement of corresponding hair lengths and angles:

$$\text{HSM} = 1 - \frac{\sum_{i=1}^N |L_i^{\text{left}} - L_i^{\text{right}}|}{\sum_{i=1}^N (L_i^{\text{left}} + L_i^{\text{right}})} \quad (34)$$

where L_i^{left} and L_i^{right} represent corresponding hair lengths on left and right sides of the head at measurement point i .

C. Aesthetic Similarity Metric (ASM)

The ASM quantifies aesthetic quality through combination of perceptual similarity measures and learned aesthetic preferences:

$$\text{ASM}(I, I^*) = \alpha \cdot \text{LPIPS}(I, I^*) + \beta \cdot \text{CNN}_{\text{aesthetic}}(I) + \gamma \cdot \text{Style}_{\text{sim}}(I, I^*) \quad (35)$$

where I is the achieved result, I^* is the target style, LPIPS is the Learned Perceptual Image Patch Similarity, and $\text{CNN}_{\text{aesthetic}}$ is a trained aesthetic quality network.

D. Safety Risk Metric (SRM)

The SRM provides comprehensive risk assessment through probabilistic modeling of potential safety events:

$$\text{SRM} = 1 - \prod_{j=1}^M (1 - P_j(\text{risk event})) \quad (36)$$

where P_j represents the probability of different safety scenarios including tool contact, excessive force, and collision events.

E. Comprehensive Evaluation Framework

TABLE IV presents our complete evaluation framework with mathematical formulations, measurement units, and typical value ranges for each metric.

F. Summary of Evaluation Metrics

To provide a comprehensive overview of our evaluation framework, TABLE V summarizes all evaluation metrics developed in this section, presenting their mathematical formulations, domains, and significance for haircutting robotics assessment.

TABLE IV
COMPREHENSIVE EVALUATION FRAMEWORK FOR HAIRCUTTING ROBOTICS

Metric Category	Specific Metric	Range	Measurement Method	Target Value
Motion Quality	Smoothness Q_1	[0, 1]	Jerk integral	> 0.85
	Accuracy Q_2	[0, 1]	Path deviation	> 0.90
	Safety Q_3	[0, 1]	Distance monitoring	> 0.95
	Efficiency Q_4	[0, ∞)	Time/energy ratio	> 0.80
Aesthetic Quality	Hair Symmetry (HSM)	[0, 1]	Length comparison	> 0.92
	Style Similarity (ASM)	[0, 1]	Perceptual networks	> 0.85
	Overall Appeal	[1, 10]	Human evaluation	> 7.5
Safety Compliance	Collision Avoidance	[0, 1]	Contact detection	= 1.0
	Force Limits	[0, 1]	Force monitoring	= 1.0
	Emergency Response	[0, 1]	Reaction time	< 0.1 s
	Risk Assessment (SRM)	[0, 1]	Probabilistic model	< 0.05
User Experience	Comfort Level	[1, 10]	Survey responses	> 8.0
	Trust Rating	[1, 10]	Behavioral measures	> 7.0
	Satisfaction Score	[1, 10]	Post-service evaluation	> 8.5

TABLE V
SUMMARY OF EVALUATION METRICS FOR HAIRCUTTING ROBOTICS

Metric	Mathematical Formulation	Domain	Significance
HMQEM	$HMQEM(T, G) = \sum_{k=1}^K w_k \cdot Q_k(T, G)$	[0, 1]	Overall motion quality assessment
Smoothness Q_1	$Q_1 = 1 - \frac{\int_0^T \ \ddot{\mathbf{x}}(t)\ ^2 dt}{\int_0^T \ \ddot{\mathbf{x}}_{ref}(t)\ ^2 dt}$	[0, 1]	Trajectory continuity and comfort
Accuracy Q_2	$Q_2 = \exp\left(-\frac{1}{N} \sum_{i=1}^N \ x_i - x_i^*\ ^2\right)$	[0, 1]	Path following precision
Safety Q_3	$Q_3 = \exp(-\lambda \max(0, d_{safe} - d_{min}))$	[0, 1]	Risk factor assessment
Efficiency Q_4	$Q_4 = \frac{T_{optimal}}{T_{actual}} \cdot \frac{E_{optimal}}{E_{actual}}$	[0, ∞)	Task completion effectiveness
HSM	$HSM = 1 - \frac{\sum_{i=1}^N L_i^{left} - L_i^{right} }{\sum_{i=1}^N (L_i^{left} + L_i^{right})}$	[0, 1]	Bilateral hair symmetry
ASM	$ASM(I, I^*) = \alpha \cdot LIPS(I, I^*) + \beta \cdot CNN_{aesthetic}(I) + \gamma \cdot Style_{sim}(I, I^*)$	[0, 1]	Aesthetic quality assessment
SRM	$SRM = 1 - \prod_{j=1}^M (1 - P_j(\text{risk event}))$	[0, 1]	Comprehensive safety risk

VII. RESEARCH CHALLENGES AND OPEN PROBLEMS

The development of VLA-based haircutting robots presents numerous research challenges that span multiple disciplines and require innovative solutions. This section identifies critical open problems and outlines promising research directions for advancing the field.

A. Perception and Sensing Challenges

Deformable Hair Modeling: Current computer vision systems struggle with the complex dynamics of hair structures that undergo continuous deformation during cutting operations. Hair exhibits non-rigid behavior with complex inter-strand interactions that challenge traditional tracking algorithms [3], [4]. Research opportunities include developing physics-informed neural networks that can predict hair behavior under cutting forces, multi-modal sensing approaches that combine vision with tactile feedback for robust hair state estimation, and real-time hair simulation methods that can support interactive planning and control.

Multi-Modal Sensor Fusion: Effective haircutting requires integration of visual, tactile, force, and auditory sensing modalities to achieve comprehensive situational awareness.

Current fusion approaches struggle with the heterogeneous nature of sensor data and the need for real-time processing with low latency [34], [35]. Critical research directions include developing adaptive fusion algorithms that can dynamically weight different sensor modalities based on task context and environmental conditions, uncertainty-aware fusion methods that properly propagate sensor noise and calibration errors, and novel sensor designs specifically optimized for hair manipulation tasks.

Aesthetic Perception: The quantification of aesthetic quality remains a fundamental challenge that requires advances in computer vision, perceptual psychology, and machine learning. Current approaches to aesthetic evaluation rely primarily on low-level image features that may not capture the complex perceptual and cultural factors that influence aesthetic judgment [9], [10]. Research opportunities include developing multi-cultural aesthetic models that account for diverse beauty standards and preferences, learning personalized aesthetic preferences from limited interaction data, and integrating temporal aesthetic evaluation that considers how hairstyles appear and evolve over time.

B. Planning and Control Challenges

Hierarchical Planning: Haircutting requires sophisticated planning capabilities that span multiple temporal and spatial scales, from high-level style planning to detailed cutting trajectories. Current planning approaches struggle with the complex interdependencies between cutting actions and the need for adaptive replanning based on intermediate outcomes [5], [6]. Key research directions include developing hierarchical reinforcement learning approaches that can effectively decompose haircutting tasks into manageable subproblems, causal reasoning methods that can predict the long-term effects of individual cutting actions, and adaptive planning algorithms that can modify strategies based on real-time feedback and changing conditions.

Safety-Critical Control: The operation of sharp cutting tools in close proximity to sensitive human anatomy requires unprecedented levels of safety assurance and fail-safe mechanisms. Traditional safety approaches based on static safety zones are insufficient for the dynamic nature of haircutting operations [7], [8]. Critical research needs include developing predictive safety models that can anticipate and prevent dangerous situations before they occur, real-time human motion prediction algorithms that enable proactive safety measures, and formal verification methods for safety-critical robotic systems operating in close human proximity.

Multiple safety risk sources have been identified including sharp tools, excessive robot forces, sensor failures, and cybersecurity vulnerabilities [1]. In teleoperated systems, network latency introduces additional hazards-if cutting tools approach too close to the scalp, remote operators may perceive events only after potentially dangerous delays. Safety mitigation strategies include network condition evaluation before task authorization, sensor redundancy (both homogeneous and heterogeneous), and obstacle-avoidance algorithms combined with soft material coverings for compliance. The selection of inherently safer tools, such as clippers over scissors, significantly reduces cut risks while maintaining functionality.

Fig. 5 illustrates the identified limitations in haircutting robot control, showing the complete process from vision input through trajectory extraction, limitation learning, inverse kinematics, and final robot action.

C. Human-Robot Interaction Challenges

Trust and Acceptance: The intimate nature of haircutting services creates unique challenges for trust establishment and user acceptance that differ significantly from other robotic applications [36], [37]. Research directions include understanding the psychological factors that influence trust in haircutting robots, developing adaptive interaction strategies that can build confidence through demonstration and gradual autonomy increase, and creating transparent AI systems that can explain their actions and reasoning to build user understanding and trust.

Cultural Sensitivity: Haircutting practices vary significantly across different cultures, ethnicities, and social contexts, requiring robotic systems that can adapt to diverse requirements and preferences [38], [39]. Key challenges include

developing cross-cultural models of aesthetic preferences and styling practices, creating adaptive interaction protocols that respect cultural norms and individual preferences, and ensuring inclusive design that serves diverse populations effectively.

D. Technical Implementation Challenges

Teleoperated vs. Autonomous Systems: Haircutting robots can be categorized into two primary types: teleoperated and fully autonomous systems [1]. Teleoperated systems are more technically feasible with current technology but face challenges from communication latencies. Long-distance teleoperation introduces fundamental limitations-the speed-of-light propagation delay between distant locations creates minimum round-trip delays of approximately 0.076 seconds, with practical internet-based systems experiencing significantly longer delays due to network infrastructure [1].

Direct Drive Technology Integration: The integration of direct drive technology represents a significant advancement for haircutting robotics [1]. Unlike conventional motors requiring large reduction ratios, direct drive systems provide high torque output with minimal reduction, enabling rapid and precise movements with high cutoff frequencies. This translates to improved responsiveness and smoother control of cutting tools, crucial for following complex head contours. The reduced mechanical backlash and increased system stiffness improve both translational and rotational movement accuracy, enabling intricate haircuts with minimal error.

Real-Time Performance: VLA models for haircutting must operate with strict latency requirements to ensure safety and user comfort. Current transformer-based architectures often struggle to meet real-time constraints while maintaining high accuracy [40], [41]. Research opportunities include developing efficient neural architectures optimized for real-time inference, edge computing approaches that can distribute processing across multiple devices, and novel training methods that can achieve high performance with reduced computational complexity.

Dataset Creation and Learning: The development of effective VLA models requires large-scale datasets of haircutting demonstrations with multi-modal annotations. Current datasets are limited in scale and diversity, hampering the development of robust learning algorithms [42], [43]. Critical needs include establishing standardized data collection protocols for haircutting demonstrations, developing privacy-preserving approaches for collecting and sharing personal grooming data, and creating simulation environments that can generate synthetic training data to augment real demonstrations.

VIII. FUTURE RESEARCH DIRECTIONS AND IMPLEMENTATION ROADMAP

The advancement of VLA-based haircutting robotics requires a coordinated research effort that addresses fundamental challenges while building toward practical implementation. This section outlines a comprehensive research roadmap with specific milestones and deliverables.

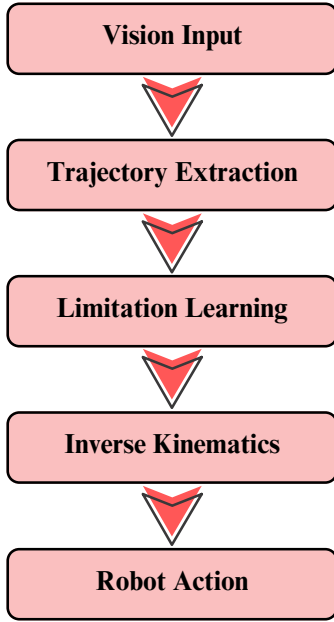


Fig. 5. Identified limitations in haircutting robot control, showing the process from vision input through trajectory extraction, limitation learning, inverse kinematics, and final robot action.

A. Near-Term Research Priorities (1-2 Years)

Foundational Sensing Capabilities: Develop specialized sensing technologies for hair perception including multi-spectral depth cameras optimized for hair geometry reconstruction, high-resolution tactile sensors for hair texture assessment, and integrated sensor suites that combine vision, force, and acoustic sensing for comprehensive hair state estimation.

Safety System Development: Create verified safety architectures including real-time collision avoidance systems with formal safety guarantees, predictive human motion models that enable proactive safety measures, and fail-safe mechanisms with hardware-software integration for emergency situations.

Basic Cutting Operations: Demonstrate fundamental cutting capabilities including straight cutting operations with length control, basic trimming and shaping functions, and integration of multiple cutting tools with automatic tool changing capabilities.

B. Medium-Term Objectives (3-5 Years)

Advanced VLA Integration: Develop sophisticated language understanding capabilities for complex styling instructions, multi-modal reasoning systems that can integrate visual, linguistic, and haptic information effectively, and adaptive learning systems that can improve performance through experience and user feedback.

Supervised Autonomy Framework: While full autonomy remains challenging, a supervised autonomy model enables early commercialization. This approach allows one human operator to supervise multiple robotic units simultaneously, intervening only when unexpected events or deviations occur. Key intervention scenarios include trajectory deviations during teleoperation, safety-critical sensor alerts, obstacle detection requiring path replanning, and explicit human requests for

intervention. This framework provides a practical path toward commercialization while maintaining safety and quality standards.

Aesthetic Reasoning: Create computational models of aesthetic quality that correlate with human aesthetic judgments, personalization algorithms that can adapt to individual style preferences, and cross-cultural aesthetic frameworks that respect diverse beauty standards and practices.

Human-Robot Collaboration: Establish effective frameworks for human-robot collaboration in haircutting including shared autonomy systems that enable seamless cooperation between human stylists and robots, natural communication interfaces for intuitive interaction, and adaptive automation that can adjust autonomy levels based on task complexity and user preferences.

C. Long-Term Vision (5-10 Years)

Fully Autonomous Systems: Develop comprehensive haircutting robots that can provide professional-quality services with minimal human supervision including complete style consultation and recommendation systems, autonomous quality assessment and correction capabilities, and integration with salon management and scheduling systems.

Advanced AI Capabilities: Integrate cutting-edge AI technologies including large-scale multi-modal models with haircutting-specific fine-tuning, creative AI systems that can generate novel hairstyle designs, and sophisticated reasoning capabilities that enable complex problem-solving and adaptation.

Commercial Deployment: Establish viable commercial products including cost-effective manufacturing approaches for haircutting robot systems, comprehensive training and certification programs for operators and technicians, and regulatory frameworks that ensure safety and quality standards.

D. Success Metrics and Evaluation

TABLE VI defines specific success metrics for each phase of the research roadmap, providing measurable objectives that can track progress toward practical implementation.

IX. BUSINESS MODELS AND COMMERCIAL IMPLEMENTATION

The commercialization of VLA-based haircutting robots presents multiple viable business models that span direct service provision, hardware manufacturing, and platform-based operations. Several key approaches have been identified for market entry and scalable deployment.

Direct Service Model: The most straightforward approach involves deploying robotic salons that operate autonomously or under minimal supervision. These systems can provide consistent, high-quality haircuts while reducing labor costs through deployment in urban centers, shopping malls, airports, and corporate facilities. The supervised autonomy model enables one operator to oversee multiple robotic units, dramatically improving labor efficiency compared to traditional one-barber-per-customer setups.

TABLE VI
RESEARCH ROADMAP SUCCESS METRICS AND TIMELINE

Research Phase	Key Milestone	Target Timeline	Success Criteria
Near-Term	Safe basic cutting	6 months	Zero safety incidents in 100 trials
	Hair perception system	12 months	95% accuracy in hair segmentation
	Multi-modal fusion	18 months	Real-time processing at 30 Hz
	Prototype demonstration	24 months	Simple haircuts in controlled environment
Medium-Term	Language understanding	36 months	90% accuracy on styling instructions
	Aesthetic evaluation	42 months	0.8 correlation with human ratings
	Style following	48 months	85% success on standard hairstyles
	Human collaboration	60 months	Effective human-robot teamwork
Long-Term	Autonomous operation	72 months	Professional-quality results
	Commercial viability	84 months	Cost-effective deployment
	Regulatory approval	96 months	Safety certification achieved
	Market deployment	120 months	Commercial product launch

Teleoperated Platform Model: A promising business model involves creating platforms where licensed barbers can offer remote services, similar to ride-sharing applications. Remote barbers can connect with customers through digital platforms, performing haircutting tasks via robotic systems from virtually anywhere. This approach expands the reach of skilled barbers while enabling flexible work arrangements across geographic regions.

Integrated Ecosystem Approach: Beyond direct services, the platform can function as a comprehensive information hub aggregating customer reviews, hairstyling trends, and product recommendations. It can serve as a marketing channel for haircare products and training courses while fostering social interactions through styling tips, tutorials, and AI-generated hairstyle suggestions. This multi-layered ecosystem encompasses commerce, communication, and community engagement.

Revenue Stream Diversification: Commercial deployment can generate diverse revenue streams including hardware sales, maintenance services, teleoperated services, software licensing, and digital platform operations. The combination of robotic capabilities with platform-based business strategies enables higher efficiency, wider reach, and enhanced customer experiences while creating new forms of value creation in both physical and digital domains.

X. CONCLUSION AND CALL TO ACTION

This position paper has established VLA-based haircutting robotics as a critical frontier for advancing embodied AI systems that can operate safely and effectively in intimate human-robot interaction scenarios. Through comprehensive analysis of existing VLA capabilities, formal problem formulations, and detailed architectural frameworks, we have demonstrated that haircutting presents unique challenges that require specialized approaches extending beyond current general manipulation systems.

Our analysis reveals that while VLA models have achieved remarkable success across various domains—with systems like RT-2 achieving 90% success rates on seen tasks and 70%

on novel scenarios—the safety-critical, aesthetic, and culturally sensitive nature of haircutting demands fundamental advances in perception, planning, and human-robot interaction. The intimate trust required for haircutting services, combined with the need for sharp tool manipulation near sensitive anatomy, positions this domain as an ideal testbed for developing next-generation safety-critical robotic systems.

The formal problem formulations introduced in this work, including the Haircutting VLA Problem and specialized evaluation metrics (HMQEM, HSM, ASM, SRM), provide mathematical foundations that enable rigorous algorithm development and performance comparison. Our proposed architectural frameworks integrate multi-modal sensing, real-time safety monitoring, and aesthetic reasoning capabilities while building upon successful VLA designs from related domains.

The comprehensive evaluation framework addresses a critical gap in robotics research by providing standardized metrics that capture the multi-faceted nature of haircutting success. Unlike traditional manipulation tasks that rely primarily on completion rates and positional accuracy, haircutting requires evaluation across technical performance, aesthetic quality, safety compliance, and user satisfaction dimensions. The metrics introduced here establish benchmarks that can drive progress and enable meaningful comparison of different approaches.

The identification of key research challenges—spanning perception of deformable objects, aesthetic reasoning, cultural sensitivity, and safety verification—provides a roadmap for coordinated research efforts across multiple institutions and disciplines. The near-term focus on foundational sensing and safety capabilities, medium-term development of advanced VLA integration and aesthetic reasoning, and long-term vision of fully autonomous commercial systems offers a structured approach to achieving practical haircutting robots.

Call to the Research Community: We call upon the robotics, AI, and human-computer interaction communities to recognize haircutting robotics as an important research area that can drive advances with broad impact. The challenges identified here—particularly in safety-critical manipulation, aesthetic reasoning, and intimate human-robot interaction—are

relevant to numerous other applications including medical robotics, personal care assistance, and creative AI systems.

Recommended Actions: Research institutions should establish dedicated haircutting robotics programs with interdisciplinary teams spanning robotics, computer vision, human factors, and aesthetic theory. Funding agencies should prioritize grants that address the fundamental challenges identified in this paper. Industry partners should collaborate on dataset creation, hardware development, and validation studies that bridge the gap between research and practical deployment.

Broader Impact: The successful development of haircutting robots has potential to transform not only personal grooming services but also our understanding of how AI systems can operate safely and effectively in intimate human contexts. The advances required for haircutting-including sophisticated aesthetic reasoning, cultural sensitivity, and safety assurance-will benefit numerous other applications where robots must understand and respond to human preferences, cultural norms, and subjective quality measures.

The framework, definitions, and research directions presented in this position paper provide a foundation for sustained research efforts that can establish haircutting robotics as a mature field with practical impact. By addressing these challenges systematically, we can develop AI systems that enhance human wellbeing through personalized, safe, and culturally sensitive service capabilities that represent a significant step toward more capable and beneficial embodied AI systems.

The time is ripe for coordinated action on this emerging research frontier. The foundational technologies exist, the theoretical frameworks are established, and the potential impact is substantial. We invite the research community to join us in advancing this exciting and important domain that promises to push the boundaries of what AI systems can achieve in service of human needs and preferences.

Author Contributions A. T. Khan conceived the study, designed the methodology, and drafted the manuscript.

S. Li supervised the project, provided critical revisions, and approved the final version. Both authors read and approved the final manuscript.

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